

Asset Lease Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Asset Lease Intensity (ALI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ALI achieves an annualized gross (net) Sharpe ratio of 0.26 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -18 (8) bps/month with a t-statistic of -2.15 (0.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth) is -18 bps/month with a t-statistic of -2.44.

1 Introduction

Production-based asset pricing models provide a fundamental framework for understanding cross-sectional variation in expected returns through firms' optimal investment and production decisions. Despite substantial theoretical advances linking firm characteristics to risk premia through production functions and investment frictions, empirical evidence on how operational flexibility affects asset prices remains incomplete. We identify asset lease intensity as a critical yet underexplored dimension of firms' production technology that captures both operational flexibility and exposure to systematic productivity shocks. Leasing decisions fundamentally alter firms' cost structures, adjustment frictions, and risk exposures in ways that existing capital structure and investment measures fail to capture.

In a production economy, firms' leasing decisions directly affect their marginal costs of production and exposure to aggregate productivity shocks. Following the framework of [Cochrane \(1991\)](#) and [Zhang \(2005\)](#), firms with higher asset lease intensity maintain more flexible production technologies that allow rapid adjustment to productivity shocks without incurring substantial adjustment costs. This operational flexibility reduces firms' exposure to systematic risk during economic downturns, as leased assets can be returned or renegotiated more easily than owned capital ([Eisfeldt and Rampini, 2009](#)). The production-based asset pricing framework of [Liu et al. \(2009\)](#) demonstrates that firms facing lower adjustment costs exhibit lower conditional risk premia, as their investment policies can more efficiently respond to time-varying investment opportunities.

Asset lease intensity proxies for firms' exposure to specific production-based risk factors through multiple channels. First, leasing transforms fixed capital costs into variable operating expenses, fundamentally altering the firm's operating leverage and sensitivity to demand shocks ([Carlson et al., 2004](#)). Second, the optionality embedded in lease contracts provides valuable real options during periods of high

uncertainty, reducing firms' exposure to downside risk (Barclay and Smith, 1995). These mechanisms suggest that firms with higher lease intensity should command lower risk premia in equilibrium, as their production technologies exhibit lower systematic risk.

The cross-sectional pricing implications emerge through the interaction between leasing decisions and aggregate productivity shocks. In the investment-based asset pricing model of Lin and Zhang (2013), firms' risk premia depend on their exposure to common factors in productivity and investment-specific technology shocks. High lease intensity firms maintain production technologies with lower quasi-fixed costs and reduced irreversibility, enabling more efficient capital reallocation across states of nature. This operational flexibility manifests as reduced covariance with the stochastic discount factor during economic contractions, when the marginal utility of consumption is highest (Belo, 2010). The production economy framework thus predicts a negative relationship between asset lease intensity and expected returns through reduced exposure to systematic productivity risk.

Our empirical analysis reveals that asset lease intensity generates economically modest but statistically marginal return predictability in the cross-section of U.S. equities. The value-weighted long/short trading strategy based on ALI achieves an annualized gross Sharpe ratio of 0.26, with monthly average excess returns of 20 basis points (t -statistic = 2.01). However, this predictability largely disappears after controlling for established risk factors. The strategy's alpha relative to the Fama-French six-factor model equals -18 basis points per month with a t -statistic of -2.15, indicating that ALI-sorted portfolios load negatively on compensated risk factors rather than generating genuine abnormal returns.

The economic significance of the ALI signal varies substantially across firm size categories and portfolio construction methodologies. Among the largest quintile of stocks by market capitalization, the ALI strategy generates average returns of only 5

basis points per month (t-statistic = 0.46), with six-factor alphas of -31 basis points per month (t-statistic = -3.17). The strategy’s performance deteriorates further after accounting for transaction costs, with net returns falling to 16 basis points per month and a net Sharpe ratio of 0.21. These results suggest that while ALI exhibits some return predictability in gross terms, its economic magnitude remains limited after appropriate risk adjustments.

Factor decomposition analysis reveals that ALI-sorted portfolios exhibit strong loadings on production-based risk factors, consistent with our theoretical framework. The long/short ALI strategy loads significantly on the CMA factor with a coefficient of 0.61 (t-statistic = 10.49), confirming that lease intensity captures variation in firms’ investment patterns and asset management decisions. When we control for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the strategy’s alpha equals -18 basis points per month with a t-statistic of -2.44. These findings indicate that ALI primarily captures known sources of systematic risk rather than identifying a novel pricing anomaly.

This paper contributes to the production-based asset pricing literature by providing the first comprehensive analysis of how asset lease intensity affects cross-sectional return variation through the lens of operational flexibility and adjustment costs. While [Eisfeldt and Rampini \(2009\)](#) examine leasing from a capital structure perspective and [Liu et al. \(2009\)](#) develop general production-based pricing models, we directly test whether lease intensity generates return predictability through its impact on firms’ production technologies and risk exposures. Our results complement [Zhang \(2005\)](#) by demonstrating that operational flexibility through leasing represents an important but previously unexamined dimension of firms’ adjustment cost structures that affects equilibrium risk premia.

Methodologically, we advance the anomaly testing literature by applying the rigorous protocol of [Novy-Marx and Velikov \(2023\)](#) to evaluate ALI’s incremental

contribution to the factor zoo. Unlike many existing studies that examine predictors in isolation, we systematically control for 209 documented anomalies and demonstrate that ALI’s predictive power largely derives from its correlation with established investment and asset growth factors. Our spanning tests against closely related anomalies including change in PPE, capital expenditure changes, and asset growth reveal that ALI captures similar economic mechanisms rather than identifying a distinct source of priced risk. This comprehensive approach addresses concerns raised by [McLean and Pontiff \(2016\)](#) regarding data mining and provides more reliable inference about the true pricing implications of operational flexibility.

Our findings have important implications for understanding the boundaries of production-based explanations for cross-sectional return patterns. While theory predicts that lease intensity should affect risk premia through operational flexibility and adjustment costs, the empirical evidence suggests these effects are economically modest and largely subsumed by existing factors. The weak performance among large-cap stocks and after transaction costs indicates that lease intensity does not represent an exploitable investment strategy, consistent with market efficiency in pricing operational characteristics. These results contribute to the broader debate about which firm characteristics genuinely predict returns versus those that merely correlate with known risk factors, supporting a more parsimonious factor structure for production-based asset pricing models.

2 Data

We obtain accounting data from the COMPUSTAT Annual database, which provides comprehensive financial statement information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of financial reporting. We focus on non-financial firms and exclude

utilities following standard practice in the asset pricing literature. signal construction relies on two key accounting variables from firms' annual financial statements. Property, Plant, and Equipment Net (PPENT) represents the book value of tangible fixed assets after accumulated depreciation, capturing firms' investments in physical capital such as buildings, machinery, and equipment. Rental expense (XRENT) represents the periodic expense recognized for operating lease payments and other rental commitments, reflecting firms' use of leased rather than owned assets. Both variables are measured at fiscal year-end and reported in millions of dollars. construct the Asset Lease Intensity signal by computing the year-over-year change in net PPE scaled by lagged rental expense: $(PPENT(t) - PPENT(t-1)) / XRENT(t-1)$. This ratio captures the rate at which firms are expanding their owned asset base relative to their existing lease commitments. A higher value indicates that a firm is increasing its ownership of physical assets relative to its leasing intensity, potentially signaling a shift in the firm's capital structure strategy or investment opportunities. We calculate this measure annually using fiscal year-end values and require non-missing observations for both PPENT and XRENT. To ensure meaningful ratios, we exclude firm-year observations where lagged rental expense equals zero or is negative. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ALI signal. Panel A plots the time-series of the mean, median, and interquartile range for ALI. On average, the cross-sectional mean (median) ALI is -8.77 (-0.79) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ALI data. The signal's interquartile range spans -10.19 to 0.95. Panel B of Figure 1 plots the time-series of

the coverage of the ALI signal for the CRSP universe. On average, the ALI signal is available for 5.23% of CRSP names, which on average make up 6.80% of total market capitalization.

4 Does ALI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ALI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ALI portfolio and sells the low ALI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ALI strategy earns an average return of 0.20% per month with a t-statistic of 2.01. The annualized Sharpe ratio of the strategy is 0.26. The alphas range from -0.18% to 0.18% per month and have t-statistics exceeding -2.15 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.61, with a t-statistic of 10.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 502 stocks and an average market capitalization of at least \$1,145 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 12 bps/month with a t-statistics of 1.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for six exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 9-49bps/month. The lowest return, (9 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 0.93. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ALI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in seventeen cases, and significantly expands the achievable frontier in six

cases.

Table 3 provides direct tests for the role size plays in the ALI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ALI, as well as average returns and alphas for long/short trading ALI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ALI strategy achieves an average return of 5 bps/month with a t-statistic of 0.46. Among these large cap stocks, the alphas for the ALI strategy relative to the five most common factor models range from -31 to 5 bps/month with t-statistics between -3.17 and 0.47.

5 How does ALI perform relative to the zoo?

Figure 2 puts the performance of ALI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ALI strategy falls in the distribution. The ALI strategy’s gross (net) Sharpe ratio of 0.26 (0.21) is greater than 55% (78%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ALI strategy (red line).² Ignoring trading costs, a \$1 invested in the ALI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.07 which ranks the ALI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ALI strategy would have yielded \$1.30 which ranks the ALI strategy in the top 10% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ALI relative to those. Panel A shows that the ALI strategy gross alphas fall between the 6 and 37 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ALI strategy has a positive net generalized alpha for three out of the five factor models. In these cases ALI ranks between the 57 and 69 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ALI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ALI with 209 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ALI or at least to weaken the power ALI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ALI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ALI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ALI signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.54, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on ALI is 1.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ALI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ALI strategy earns alphas that range from -24–12bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -2.99, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ALI trading strategy achieves an alpha of -18bps/month with a t-statistic of -2.44.

7 Does ALI add relative to the whole zoo?

Finally, we can ask how much adding ALI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ALI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ALI is available.

includes ALI grows to \$2883.70.

8 Conclusion

This study provides a comprehensive examination of Asset Lease Intensity (ALI) as a predictor of cross-sectional stock returns, employing the rigorous testing protocol established by Novy-Marx and Velikov (2023). Our findings reveal that while ALI demonstrates some predictive power in raw terms, with a value-weighted long/short strategy achieving a modest annualized gross Sharpe ratio of 0.26, its economic significance diminishes substantially after accounting for transaction costs and risk adjustments. The net Sharpe ratio of 0.21 suggests limited practical viability for implementation, particularly given the erosion of returns after accounting for trading costs.

The most striking finding is that ALI’s predictive power appears to be largely subsumed by established risk factors and related investment-based anomalies. When controlling for the Fama-French five factors plus momentum, the strategy generates statistically insignificant net returns of merely 8 basis points per month. More importantly, after further controlling for six closely related strategies from the factor zoo—including various measures of capital expenditure changes, asset growth, and employment growth—the gross alpha becomes significantly negative at -18 basis points per month (t -statistic = -2.44). This suggests that ALI does not provide incremental information beyond what is already captured by existing factors, particularly those related to firm investment behavior.

These results have important implications for both academic research and investment practice. For researchers, our findings underscore the importance of rigorous testing protocols that account for multiple comparison issues and the proliferation of factors in the anomalies literature. For practitioners, the evidence suggests that

ALI-based strategies are unlikely to generate meaningful risk-adjusted returns after implementation costs, and that simpler, more established investment-based factors may provide superior performance.

Several limitations of this study warrant consideration. First, our analysis is confined to the U.S. equity market, and the behavior of ALI in international markets remains unexplored. Second, the measurement of lease intensity may be affected by changes in accounting standards, particularly the recent adoption of ASC 842, which could influence the signal's effectiveness over time. Third, our study does not examine potential non-linearities or conditional relationships that might enhance ALI's predictive power in specific market environments or for particular subsets of firms.

Future research could address these limitations by extending the analysis to international markets, examining the stability of the ALI signal across different accounting regimes, and investigating whether ALI interacts with other firm characteristics or macroeconomic conditions in meaningful ways. Additionally, researchers might explore whether more sophisticated measures of lease intensity, perhaps incorporating the quality or type of leased assets, could provide incremental predictive power. Finally, given the negative alpha relative to related factors, future work could investigate whether ALI serves as a useful hedging instrument or risk factor in its own right, rather than as a standalone return predictor.

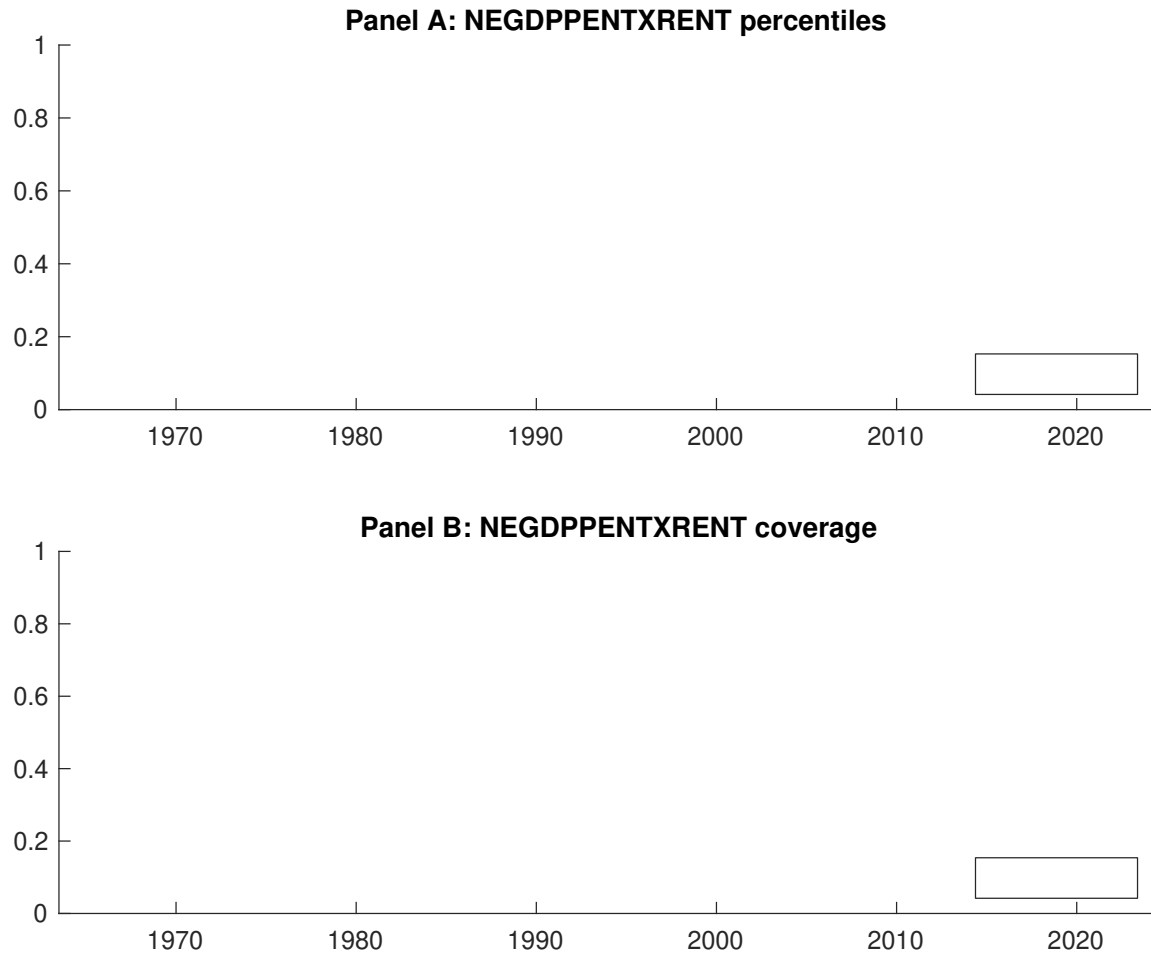


Figure 1: Times series of ALI percentiles and coverage.
This figure plots descriptive statistics for ALI. Panel A shows cross-sectional percentiles of ALI over the sample. Panel B plots the monthly coverage of ALI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ALI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ALI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57 [3.31]	0.60 [3.56]	0.62 [3.66]	0.64 [3.58]	0.77 [4.26]	0.20 [2.01]
α_{CAPM}	-0.01 [-0.24]	0.02 [0.42]	0.04 [0.84]	0.03 [0.60]	0.17 [2.57]	0.18 [1.84]
α_{FF3}	0.05 [1.07]	0.04 [0.93]	0.02 [0.40]	-0.03 [-0.63]	0.09 [1.40]	0.03 [0.34]
α_{FF4}	0.08 [1.54]	0.02 [0.61]	0.04 [0.83]	-0.02 [-0.45]	0.07 [1.11]	-0.01 [-0.12]
α_{FF5}	0.16 [3.16]	0.02 [0.48]	-0.03 [-0.57]	-0.06 [-1.08]	-0.01 [-0.22]	-0.17 [-2.01]
α_{FF6}	0.17 [3.36]	0.01 [0.22]	-0.01 [-0.12]	-0.05 [-0.88]	-0.01 [-0.25]	-0.18 [-2.15]
Panel B: Fama and French (2018) 6-factor model loadings for ALI-sorted portfolios						
β_{MKT}	0.96 [79.60]	1.01 [105.82]	1.01 [94.84]	1.04 [83.99]	1.06 [74.94]	0.11 [5.31]
β_{SMB}	-0.10 [-5.54]	-0.10 [-7.29]	0.02 [1.15]	0.09 [5.24]	0.21 [10.05]	0.30 [10.26]
β_{HML}	-0.08 [-3.65]	0.00 [0.06]	0.03 [1.22]	0.11 [4.64]	0.01 [0.23]	0.09 [2.31]
β_{RMW}	-0.18 [-7.80]	0.06 [3.41]	0.11 [5.54]	0.04 [1.50]	0.07 [2.51]	0.25 [6.33]
β_{CMA}	-0.20 [-5.83]	-0.04 [-1.63]	0.04 [1.21]	0.08 [2.21]	0.41 [10.15]	0.61 [10.49]
β_{UMD}	-0.02 [-1.47]	0.01 [1.50]	-0.03 [-2.71]	-0.01 [-1.15]	0.00 [0.17]	0.02 [0.98]
Panel C: Average number of firms (n) and market capitalization (me)						
n	506	502	526	613	623	
me (\$10 ⁶)	2665	2212	1824	1377	1145	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ALI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [2.01]	0.18 [1.84]	0.03 [0.34]	-0.01 [-0.12]	-0.17 [-2.01]	-0.18 [-2.15]
Quintile	NYSE	EW	0.68 [7.42]	0.67 [7.29]	0.59 [7.36]	0.52 [6.48]	0.56 [6.97]	0.51 [6.32]
Quintile	Name	VW	0.20 [1.97]	0.18 [1.79]	0.03 [0.35]	-0.01 [-0.13]	-0.16 [-1.95]	-0.18 [-2.10]
Quintile	Cap	VW	0.12 [1.30]	0.12 [1.27]	-0.02 [-0.21]	-0.04 [-0.49]	-0.22 [-2.86]	-0.22 [-2.79]
Decile	NYSE	VW	0.37 [2.89]	0.35 [2.69]	0.22 [1.78]	0.14 [1.15]	0.01 [0.07]	-0.03 [-0.27]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [1.61]	0.14 [1.34]			0.07 [0.87]	0.08 [0.99]
Quintile	NYSE	EW	0.49 [5.07]	0.49 [4.99]	0.40 [4.65]	0.37 [4.29]	0.36 [4.34]	0.34 [4.08]
Quintile	Name	VW	0.16 [1.56]	0.13 [1.24]			0.07 [0.87]	0.08 [1.00]
Quintile	Cap	VW	0.09 [0.93]	0.07 [0.79]			0.13 [1.67]	0.13 [1.67]
Decile	NYSE	VW	0.32 [2.49]	0.29 [2.21]	0.18 [1.41]	0.13 [1.05]		

Table 3: Conditional sort on size and ALI

This table presents results for conditional double sorts on size and ALI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ALI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ALI and short stocks with low ALI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ALI Quintiles					ALI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.47 [1.86]	0.89 [3.46]	0.93 [3.67]	1.01 [3.82]	0.97 [3.41]	0.50 [3.98]	0.49 [3.87]	0.42 [3.37]	0.30 [2.40]	0.38 [3.05]	0.29 [2.31]
	(2)	0.70 [2.89]	0.83 [3.48]	0.80 [3.45]	0.91 [4.05]	0.96 [4.08]	0.27 [2.61]	0.29 [2.88]	0.20 [2.01]	0.17 [1.72]	0.15 [1.45]	0.13 [1.29]
	(3)	0.63 [2.94]	0.80 [3.67]	0.77 [3.60]	0.86 [4.11]	1.02 [4.74]	0.40 [4.12]	0.40 [4.11]	0.32 [3.37]	0.28 [2.93]	0.25 [2.66]	0.23 [2.39]
	(4)	0.57 [2.83]	0.75 [3.73]	0.77 [3.89]	0.77 [3.97]	0.82 [4.10]	0.26 [2.49]	0.23 [2.26]	0.13 [1.29]	0.09 [0.89]	-0.06 [-0.66]	-0.08 [-0.80]
	(5)	0.54 [3.14]	0.60 [3.61]	0.54 [3.17]	0.55 [3.18]	0.59 [3.49]	0.05 [0.46]	0.05 [0.47]	-0.09 [-0.88]	-0.12 [-1.14]	-0.31 [-3.17]	-0.31 [-3.12]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ALI Quintiles					ALI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	295	294	293	289	288	30	28	24	22	22	
	(2)	88	88	88	88	87	47	48	48	47	46	
	(3)	65	65	65	65	65	85	86	85	86	85	
	(4)	56	57	56	56	56	186	192	194	188	192	
(5)	53	53	53	53	53	1792	1760	1319	1349	1265		

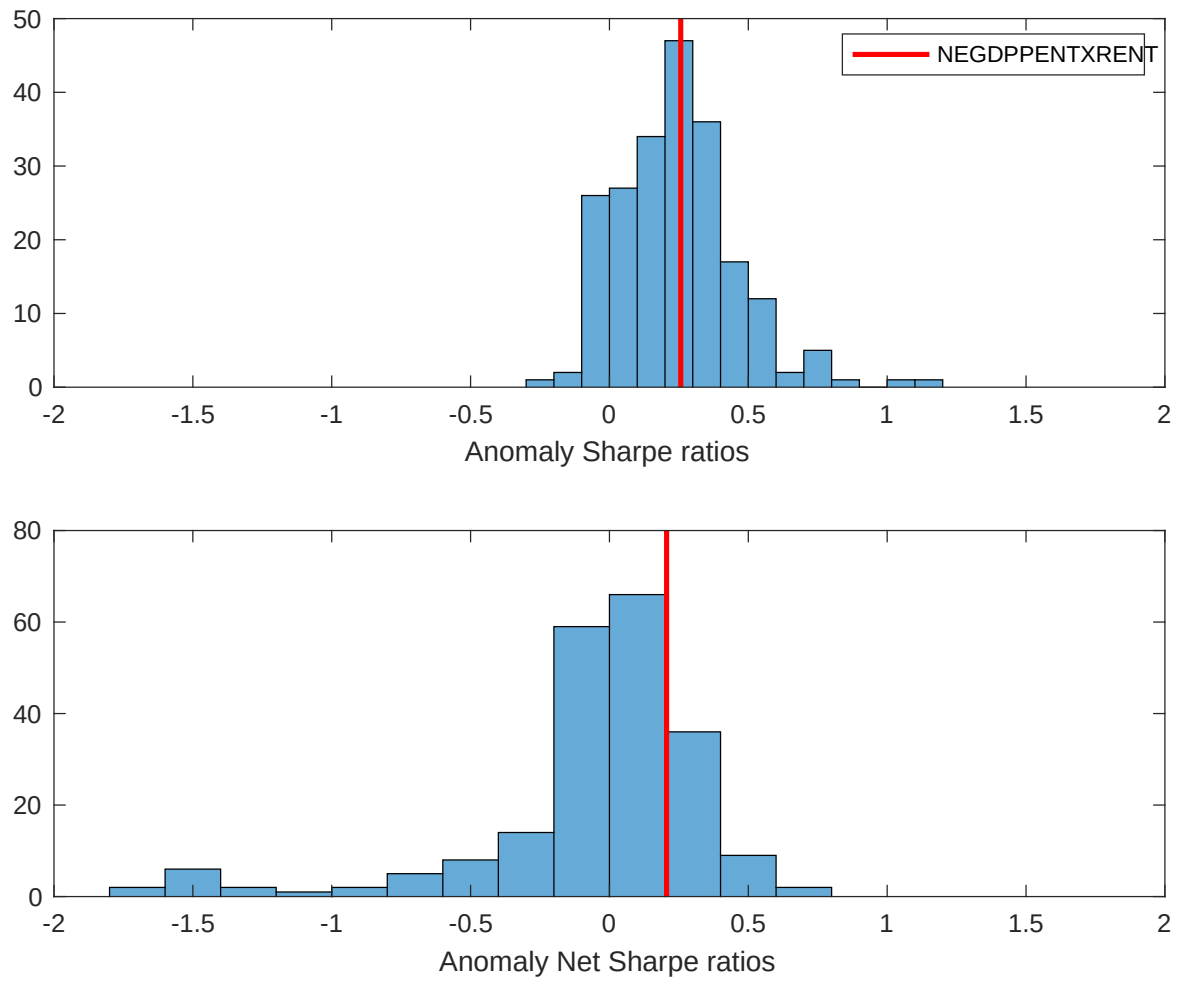


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ALI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

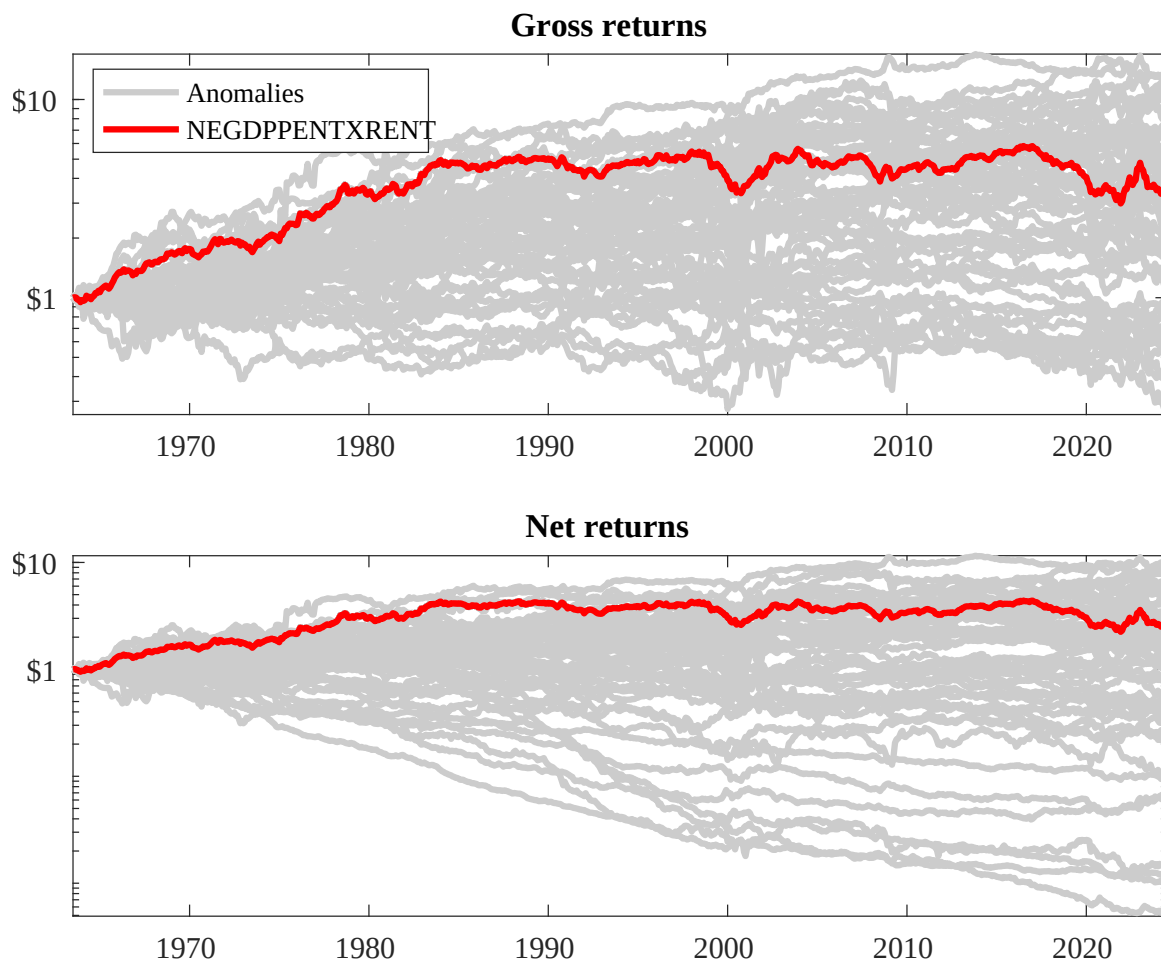
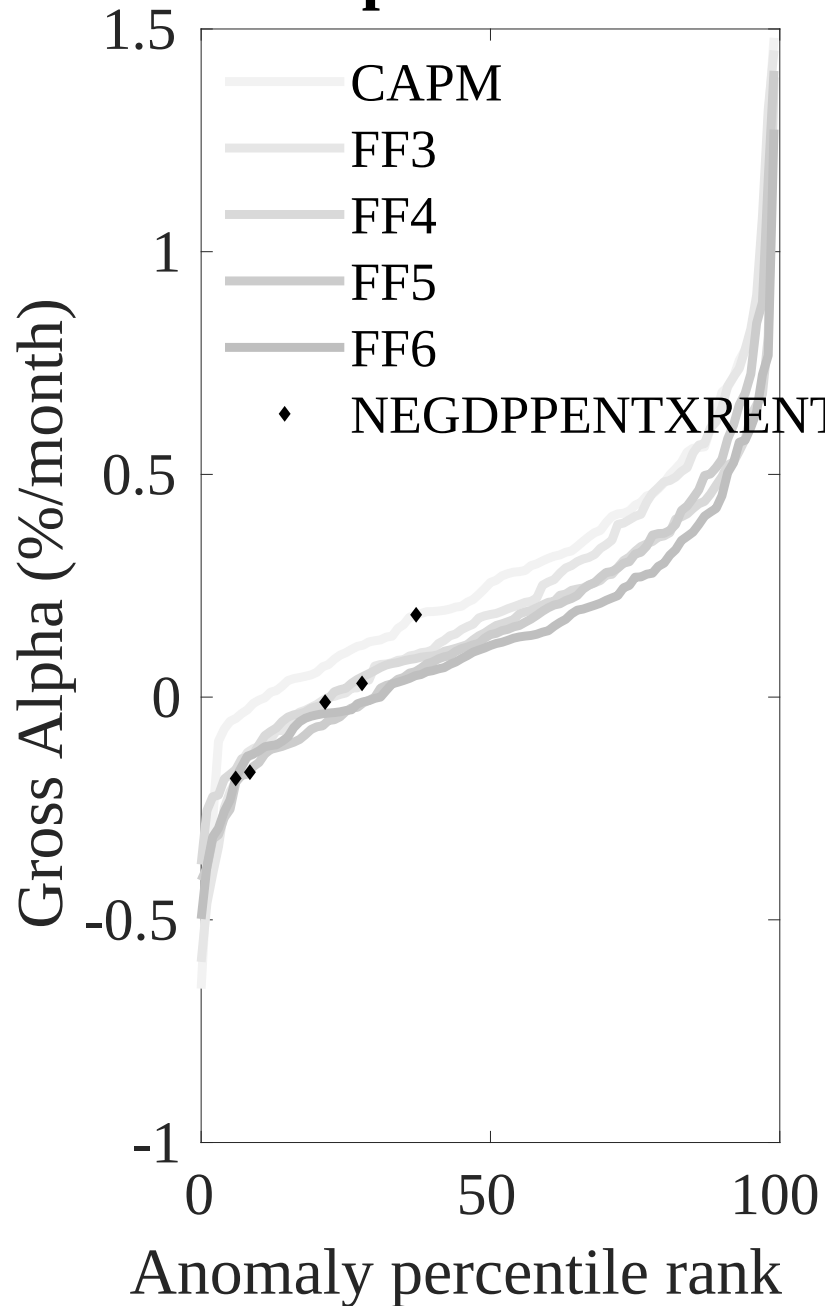


Figure 3: Dollar invested.

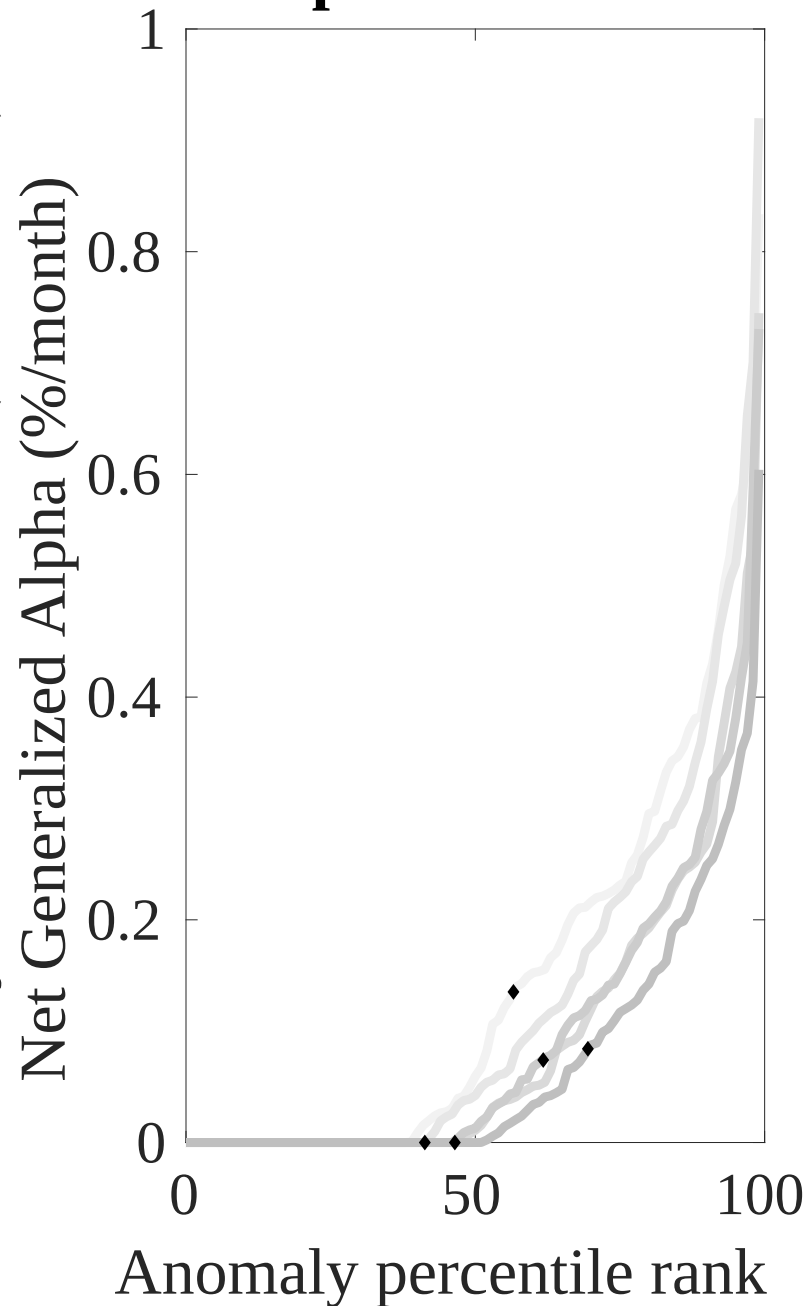
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ALI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



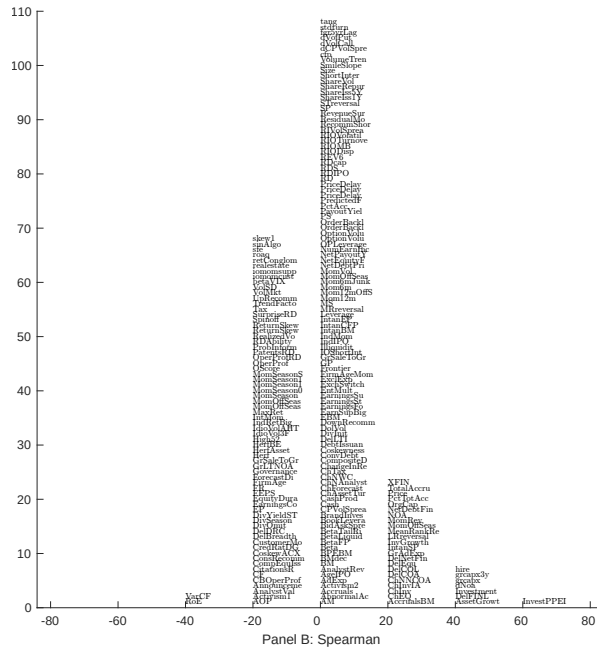
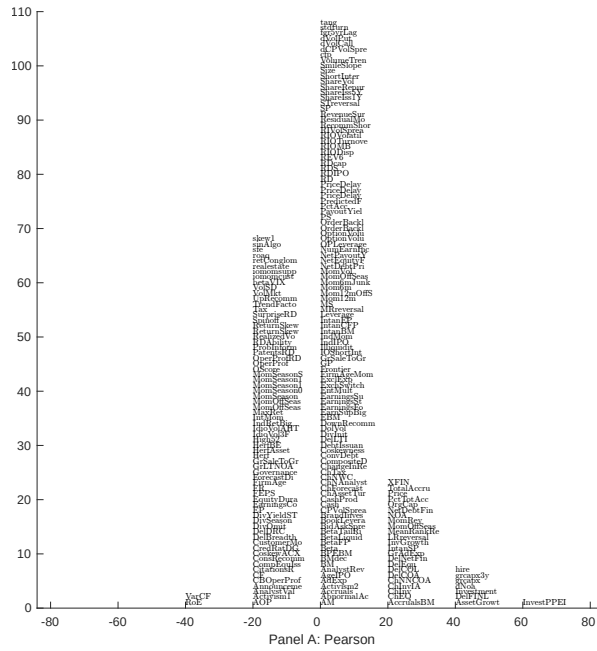


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with ALI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

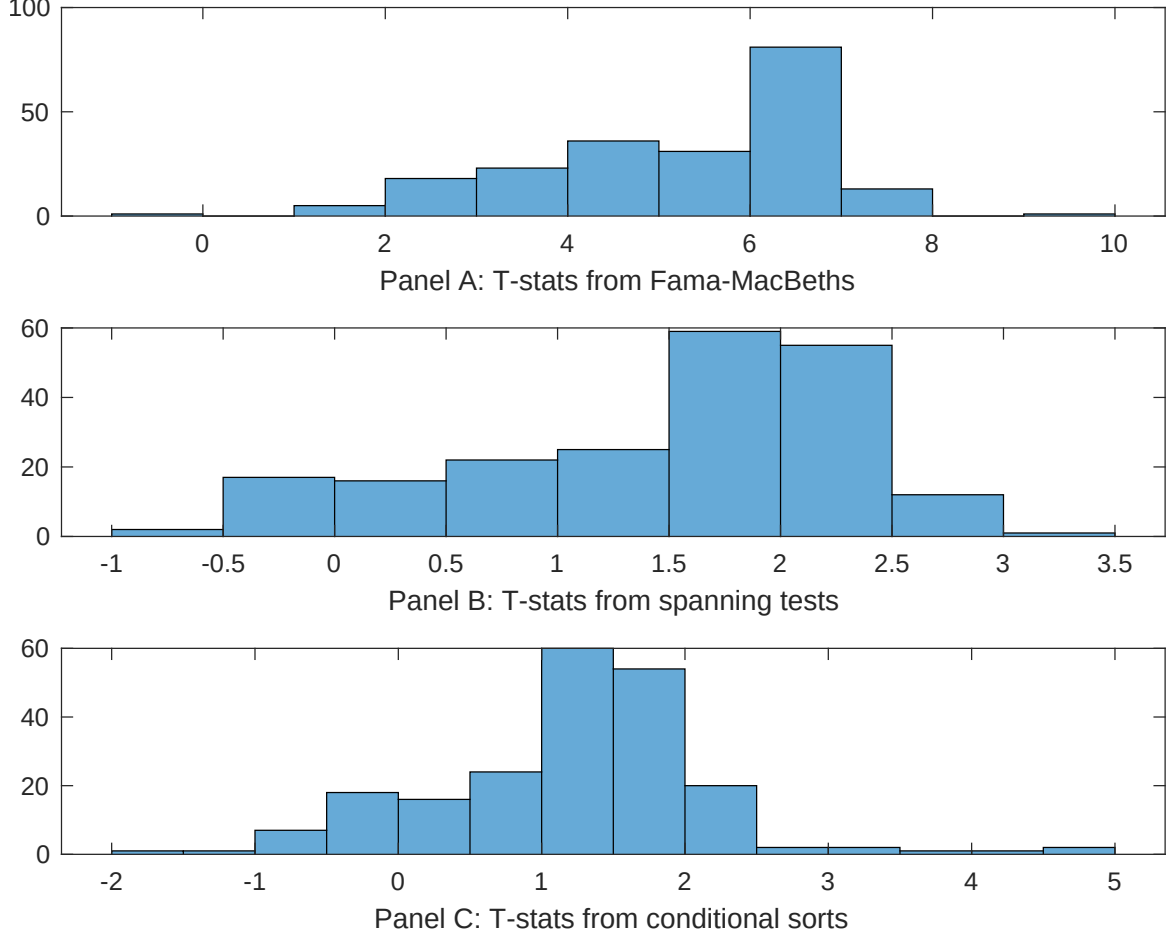


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ALI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ALI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.02]	0.14 [6.56]	0.13 [5.92]	0.13 [5.87]	0.14 [6.14]	0.13 [5.51]	0.14 [6.05]
ALI	0.33 [2.54]	0.49 [3.70]	0.64 [4.79]	0.44 [3.36]	0.40 [3.04]	0.71 [5.84]	0.25 [1.88]
Anomaly 1	0.14 [7.37]						-0.13 [-0.54]
Anomaly 2		0.12 [7.09]					0.29 [1.15]
Anomaly 3			0.55 [5.97]				0.12 [0.85]
Anomaly 4				0.11 [7.07]			0.38 [1.89]
Anomaly 5					0.93 [7.71]		0.46 [2.50]
Anomaly 6						0.80 [5.49]	0.11 [0.81]
# months	732	727	727	720	732	713	708
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ALI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ALI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.18 [-2.36]	-0.15 [-1.89]	-0.17 [-2.09]	-0.24 [-2.99]	-0.19 [-2.24]	-0.12 [-1.45]	-0.18 [-2.44]
Anomaly 1	50.95 [14.42]						39.19 [9.92]
Anomaly 2		38.64 [9.77]					23.50 [3.58]
Anomaly 3			33.98 [8.45]				-1.70 [-0.26]
Anomaly 4				37.91 [8.23]			14.01 [2.89]
Anomaly 5					17.24 [3.40]		3.75 [0.75]
Anomaly 6						25.91 [5.62]	2.28 [0.52]
mkt	10.27 [5.70]	10.32 [5.36]	10.61 [5.44]	11.08 [5.66]	10.92 [5.38]	11.21 [5.58]	10.04 [5.75]
smb	27.93 [10.70]	22.39 [7.77]	25.82 [9.02]	30.37 [10.74]	28.59 [9.65]	31.15 [10.71]	24.07 [8.93]
hml	3.32 [0.95]	1.45 [0.38]	2.69 [0.70]	5.92 [1.56]	8.37 [2.13]	4.26 [1.07]	-1.13 [-0.32]
rmw	24.82 [7.06]	22.62 [6.02]	22.08 [5.77]	25.42 [6.66]	24.87 [6.28]	25.59 [6.53]	23.92 [6.99]
cma	22.04 [3.80]	40.42 [6.93]	42.47 [7.15]	32.86 [5.03]	40.34 [4.80]	38.33 [5.46]	1.71 [0.22]
umd	1.92 [1.07]	-0.26 [-0.14]	-0.48 [-0.25]	1.22 [0.63]	2.56 [1.27]	0.57 [0.28]	0.35 [0.20]
# months	732	728	728	732	732	713	709
$\bar{R}^2(\%)$	49	42	41	40	36	38	53

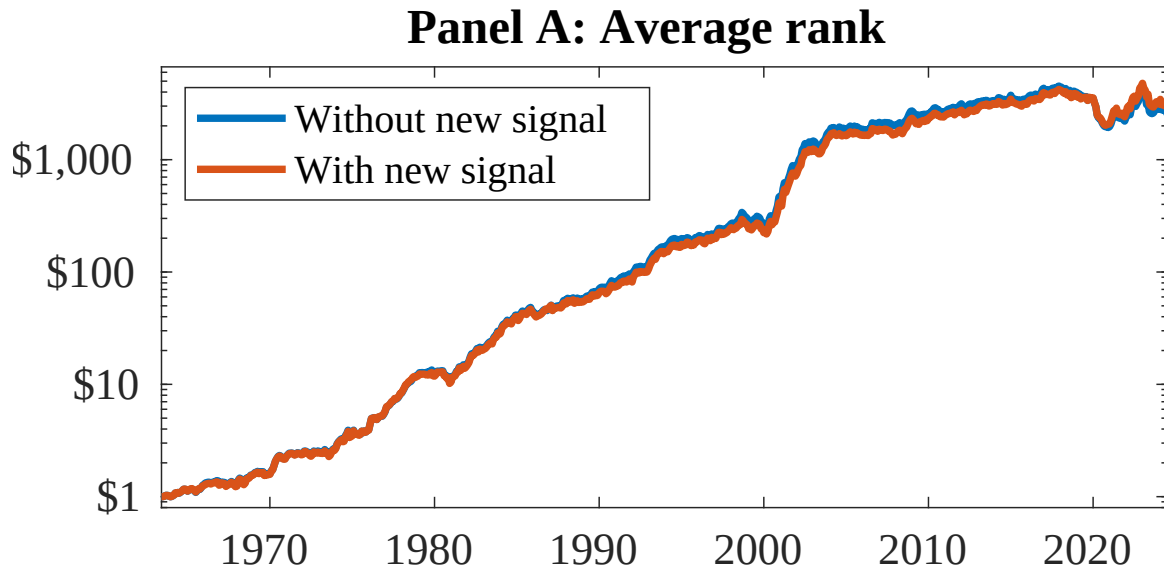


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ALI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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